

Kaustubh Patange

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EDUCATION

Walchand Institute of Technology

Bachelor of Technology in Computer Science and Engineering; CGPA: 8.98

Minor Degree in Electronics and Telecommunication Engineering;

Solapur, Maharashtra, India

Aug 2023 – April 2027

RELATED COURSEWORKS

- Linear Algebra
- Digital System Design
- Database Management Systems
- Scientific Computing
- Backend Systems
- Data Structures & Algorithm

EXPERIENCE

GSoC Mentee, Core Contributor ([Link](#))

November 2025 – Present

stdlib-js/stdlib

- Successfully contributed 100+ PRs including entire statistics and **BLAS** namespace for n-dim-arrays, core basic linear algebraic subroutines along with testing and benchmarking.
- Reviewed multiple PRs to maintain repository guidelines and suggested code fixes.
- Extending Level 2 and Level 3 **BLAS** routines for linear algebra and C addons as part of **GSoC 2026 Cohort**.

Software Developer ([Link](#))

July 2025 – September 2025

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- Developed and maintained backend services for a Warehouse Management System (WMS), implementing secure RESTful APIs to support inventory tracking, order processing, and warehouse operations.
- Designed and optimized database schemas and business logic, improving data consistency and enabling reliable management of inventory, stock movements, and fulfillment workflows while collaborating with the engineering team to deliver production-ready features.

PROJECTS

CoSketch – Real-time Collaborative Whiteboard | *Next.js, TypeScript, WebSockets, PostgreSQL* [Live](#) [Code](#)

- Engineered a low-latency real-time collaboration system using persistent WebSocket connections to synchronize canvas operations across multiple concurrent users.
- Designed an event-driven canvas state architecture for drawing, shape transformations, and persistent workspace management, ensuring consistent client synchronization and scalable collaboration.

Corpora – Multimodal Agentic RAG Platform | *Next.js, LangGraph, Qdrant, PostgreSQL, Redis* [Live](#) [Code](#)

- Engineered a multi-tenant agentic RAG pipeline using LangGraph with hybrid vector search (dense + BM25, RRF fusion) and cross-encoder reranking, streaming citation-grounded answers over SSE at 31 ms p95 retrieval latency.
- Built an async ingestion and CLIP text-to-image search system with CI-gated retrieval evals (faithfulness, hit-rate), Clerk JWT multi-tenancy, and Langfuse observability, deployed on production cloud infra.

TECHNICAL SKILLS

Languages: C/C++, Java, Python, JavaScript, TypeScript, Fortran

Frameworks: ExpressJS, NodeJS, EJS, FastAPI, ReactJS, NextJS, TailwindCSS, Bootstrap

AI & ML: PyTorch, OpenCV, Langchain, LangGraph, pgvector, Pinecone, Pandas

Other: Bash, Postman, Git, Docker, AWS, Kafka, Redis, MongoDB, Supabase, Postgres, SQL

ACHIEVEMENTS & CERTIFICATIONS

LeetCode: Solved 200+ data structure and algorithm problems. ([Profile](#))

stdlib: Member of the numerical library called stdlib on GitHub.

NPTEL Online Certification: Programming with Generative AI (Aug-Oct 2025). ([Certificate](#))

Infosys Springboard: Agile Scrum Certification (May 2026). ([Certificate](#))